Churn Analysis & Customer Intelligence

1. The Business Challenge

In the hyper-competitive OTT landscape (Netflix, Hotstar, Prime), retention is the only way to survive. You will act as a Data Analyst tasked with identifying high-risk subscribers using a multi-dimensional dataset (Customer demographics, Subscription tiers, and Support escalations).

2. Core Tech Stack

SQL & Python Integration (numpy, pandas, sqlite3, matplotlib, seaborn), Performance Engineering (data cleaning, feature engineering, analytics), Behavioral Visualization, Writing Actionable Insights

3. Project Milestones

- Relational Data Extraction: Connecting Python to SQL databases to pull multi-table datasets.
- Advanced Feature Engineering: Data imports, calculating tenure, churn rates, and customer aging.
- Executive Reporting: Translating technical findings into billion-dollar business insights.

Churn Analysis - Roadmap

- Connect SQL database to Python – pandas and sqlite3
- Data import using SQL query in Python – pandas and sqlite3
- Data Cleaning – numpy and pandas (data types, rename cols, select specific cols, QCs, handle missing/null values)
- Feature Engineering – numpy and pandas (Create new calculated cols, data transformation, use filters)
- Data Analysis – numpy and pandas (EDA - aggregation, group by, pivot table)
- Data Visualization – matplotlib and seaborn
- Insights on ppt – key insights, next steps, impact to show in resume

Churn Definition by Business

Churn analysis is the process of figuring out why customers stop doing business with you. For this, we only need to look at three things:

- "Who": Identifying which customers left or will leave in future
- "Why": Analyzing their behavior before they left.
- "When": Finding the "danger zone" – point when a user leave

Business Type	Churn Definition

SaaS	Subscription canceled
E-commerce	No purchase in 90 days
Adtech	Using service/app in 90 – 120 days
Streaming	Membership inactive
Telecom	Account terminated
Banking	No transactions for X months

Database: customer_churn Tables

db_customer	db_subscription	db_support
customerid name country State gender dob interests pincode	customerid subscription_start_date subscription_type renewal_date plan_type contract_type cancellation_date cancellation_reason monthly_charges cltv churn_score	customerid complaint_date Escalations csat_score col_1 comment

Calculated Metrics

KPI	Formula
churn rate	churned customer / total customers
Churn by plan type	churn rate GROUP BY plan_type (Basic/Standard/Premium)
Churn by state	churn rate GROUP BY country, state
Retention rate	1-churn rate
ARPU	SUM(monthly_charges) / COUNT(active customerid)
Average customer tenure	AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))
Revenue at risk	SUM(monthly_charges) WHERE churn_score > 70
Escalation rate	SUM(escalations) / COUNT(complaints) × 100

Avg complaints per customer	COUNT(complaints) / COUNT(DISTINCT customerid)
Correlation: escalations → churn	churn rate WHERE escalations ≥ 1 vs 0

Insights

- Churn Rate: 28.6% | Retention Rate: 71.4%
- Most of the churn is from basic subscription plan – nothing to worry in terms of major revenue impact
- Most of the churn happened in the month of Sep 2024 and, most affected state is Karnataka
- Average Tenure (Days): 1,451 | ARPU is Rs 18.8
- Total Revenue = 395
- Revenue loss due to churn = 74 | CLTV Lost = 2,047
- % Revenue loss = 18%
- monthly vs annual churn = 55.6% vs 8.3%

Action Items

- Check what happened in Karnataka – was there any increase in price, user complaints, tech issues, etc
- Did we increased subs price for Basic plan or recently made any changes – specially in sep month?
- Check what competitor is doing – bcoz one user has moved to competitor
- Focus on customers with ‘High’ & ‘Med’ churn risk, check their LTV (to make priority list), complaint request, reach out to them through email, sms, calls and fix their issue.

Portfolio Project - Summary

Churn & Revenue Impact (general summary)

Engineered an end-to-end churn analytics pipeline for an OTT subscription platform by integrating multi-table subscriber data across acquisition type, contract structure, and plan tier (20+ KPIs). Uncovered a significant churn disparity between monthly and annual contract segments, quantified MRR leakage and CLTV erosion attributed to high-risk cohorts, and delivered a data-backed contract-migration retention strategy to reduce involuntary subscriber loss.

Risk Scoring & Segmentation (Risk Analysis & Segmentation)

Developed a multi-dimensional churn risk scoring model by synthesising subscription tenure, plan type, and support escalation signals across three relational tables (20+ KPIs), segmented the customer base into risk tiers using composite churn scores, exposed a significant lifetime value gap between churned and retained cohorts, and recommended prioritising Premium annual-plan retention over Basic monthly acquisition to maximise long-term revenue yield.

Support Intelligence & Cancellation Analysis (working on live data)

Performed cross-functional support- churn correlation analysis by joining complaint, escalation, and CSAT data with subscription records, identified that escalated support interactions were disproportionately concentrated among churned customers, decomposed cancellation drivers into competitor switching, pricing sensitivity, and content dissatisfaction, and translated findings into a prioritised product and pricing roadmap presented to the platform's growth team.

Churn & Revenue Impact (general summary + with numbers from dataset)

Engineered an end-to-end churn analytics pipeline for an OTT subscription dataset (20+ KPIs), identifying a 28.6% overall churn rate and surfacing that monthly-contract subscribers churned at 55.6%, which is 6.7 times the 8.3% annual-contract rate- directly attributing $73.94/mo in MRR leakage and $2,047 in CLTV erosion to six at-risk customers, enabling a targeted contract-migration retention strategy.